

10. Operating Instructions, Limitations, Fuse-Replacement Information, and Shutdown/Storage Procedure

10.1 Safe Operating Procedure

Phase 1: Pre-Power Inspection & Setup

1. Inspect the main unit, acrylic enclosure, banana jacks, and cables for any visible physical damage.
2. Ensure the main **PS_ON** toggle switch is set to the OFF (Open) position.
3. Verify that all external load connections are unplugged from the output terminals prior to starting.
4. Connect the main AC power cord to the PLENTY COMPUTER (ATX500WS) AC inlet and plug it into an earthed AC wall outlet.
5. Verify that the Amber LED Indicator (+ 5VSB Standby Power) illuminates, confirming AC mains power is active.

Phase 2: System Power Enablement & Operation

1. Flip the main **PS_ON** toggle switch to the ON (Closed to **COM**) position.
2. Confirm that the source PSU cooling fan begins to spin and the Green LED Indicator illuminates, signaling active main DC rails.
3. For fixed voltage rails , the connect test leads to the respective banana terminals.
4. For the adjustable buck-boost output, set the desired output voltage and current limit using the module potentiometers while monitoring the DSN-VC288 display panel meter before connecting sensitive loads.

10.2 Operational Boundaries & System Limitations

Rail / Output Parameter	Maximum Continuous Current	Maximum Load Power
+ 3.3VDC Rail	5.0A	16.5W
+ 5.0VDC Rail	5.0A	25.0W
+ 12.0VDC Rail	5.0A	60.0W
Buck-Boost Converter	3.0A	35.0W

10.3 Branch Fuse Replacement Procedure

Step-by-Step Fuse Replacement:

1. Immediately disconnect all external loads from the output terminals.
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2. Switch **PS_ON** to OFF and unplug the AC power cable from the wall outlet.
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3. Unscrew the panel-mount fuse holder cap for the affected rail counter-clockwise.
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4. Remove the blown glass fuse and inspect for a melted element or blackened glass tube.
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5. Insert a new, identically rated WIREMAN 5×20 mm fast-acting glass fuse.
6. Re-tighten the fuse cap securely, re-connect AC power, and test the rail under no-load conditions prior to resuming operation.

10.4 Safe Shutdown & Storage Procedure

Shutdown Sequence:

1. Turn off or disconnect all connected load circuits/breadboards from the output binding posts.
2. Toggle the main **PS_ON** switch to the OFF position. Verify that the cooling fan stops and the Green Power LED turns off.
3. Unplug the AC mains cable from the power inlet. Verify that the Amber Standby LED turns off after residual energy dissipates.

Storage & Maintenance Guidelines:

1. Storage Environment: Store the unit in a clean, dry, non-condensing environment away from direct sunlight, high heat, and corrosive chemicals.
2. Cable Management: Coil the AC power cord loosely; do not bend or apply sharp tension to the output harness entry grommet.
3. Inspection Before Re-use: Perform a visual check on terminal tightness, panel clarity, and cable insulation integrity prior to the next laboratory session